
L Confounder Investigation: Data Contamination

A potential confounder in our analysis is the influence of data contamination. If prompts closely resemble the training data (including pretraining and alignment tuning, i.e., "data contamination"), smaller BF values would be expected, and vice versa. To evaluate this, we use the Min-K% metric (Shi et al., 2024b), which quantifies the overlap between experimental prompts and training data. Following Shi et al. (2024b), we set $K = 20$ and compute the average log-likelihood for the minimum $K\%$ of tokens. Using these Min-K% values, we perform a linear regression with BF to assess their correlation. For each task-model pair, Signed R^2 values are reported to indicate the strength and sign (positive or negative) of the correlation.

The results of the Min-K% analysis are presented in Figure 19. Significant negative correlations between BF and Min-K% are observed for models such as Llama-3-8B-Instruct, Llama-3-70B-Instruct, and Llama-2-70B-Chat across several tasks. Conversely, Llama-3-8B and Llama-2-13B-Chat models exhibit positive correlations. For other models, correlations are notably weaker. Overall, there is no consistent correlation pattern between BF and Min-K% across datasets and models, suggesting that data contamination cannot fully explain our findings.

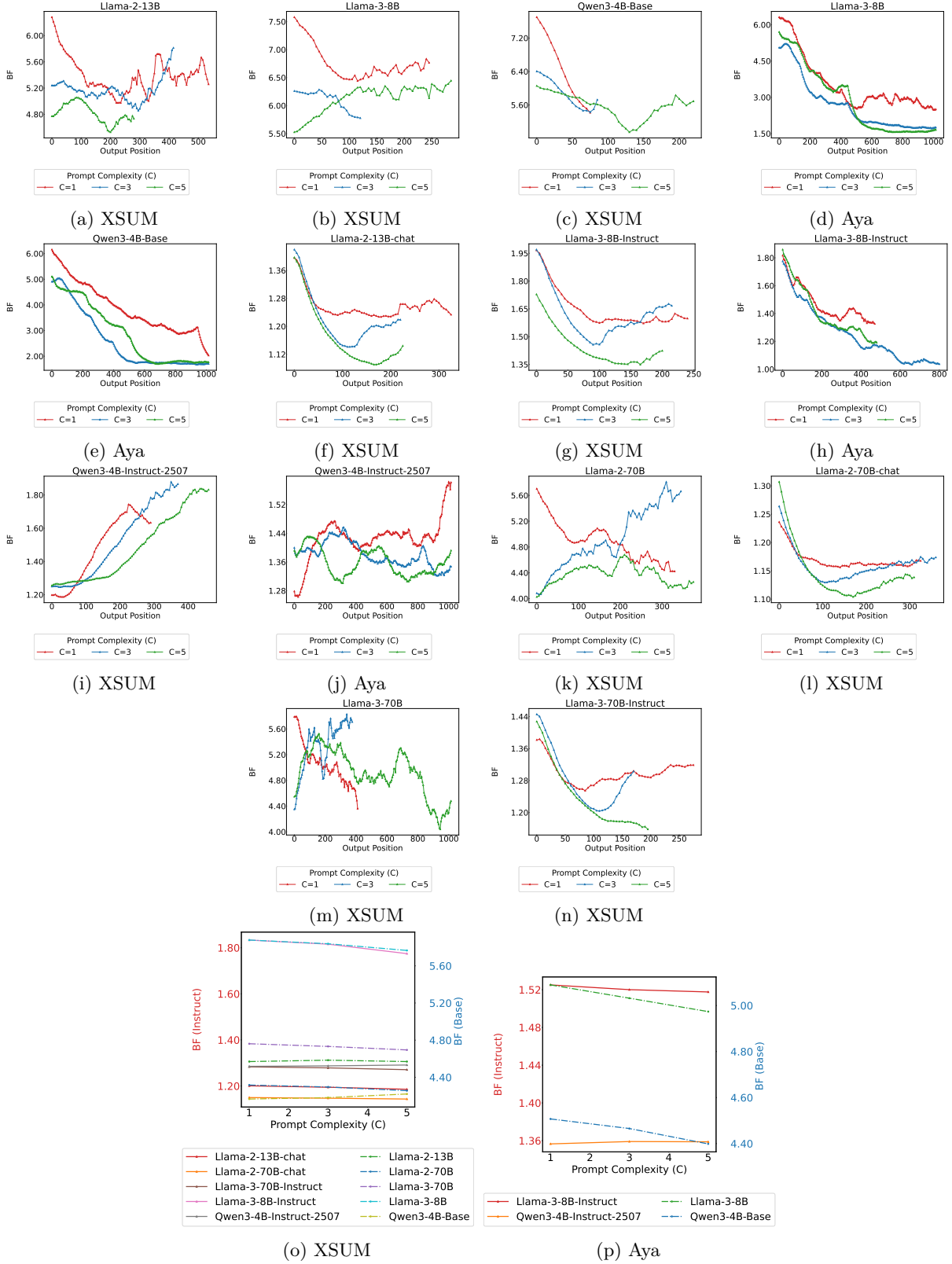


Figure 13: **Additional Verification on Summarization, Multilingual and Qwen Model family.** For better visualization, we compute the exponential moving averaged values of perplexity with the smoothing factor set as 0.1.

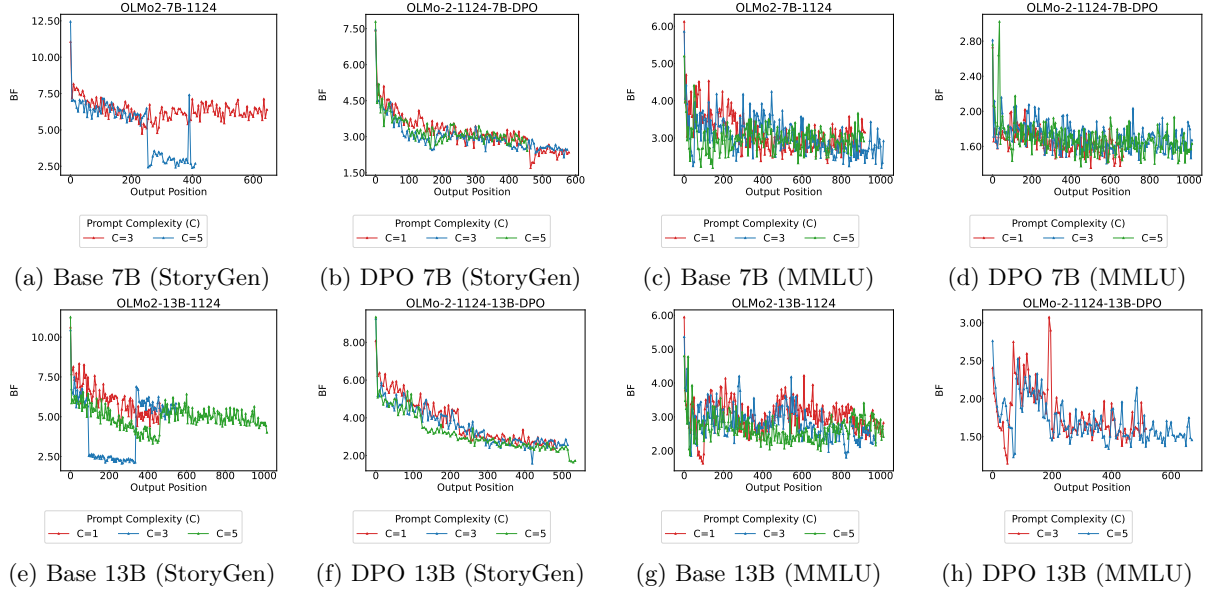


Figure 14: **BF Output Dynamic for OLMo-2 (7B & 13B) across alignment stages.** We compare Base and DPO models on Creative StoryGen and MMLU. Note that we treat DPO as the aligned model for OLMo-2, following the convention in (OLMo et al., 2024).

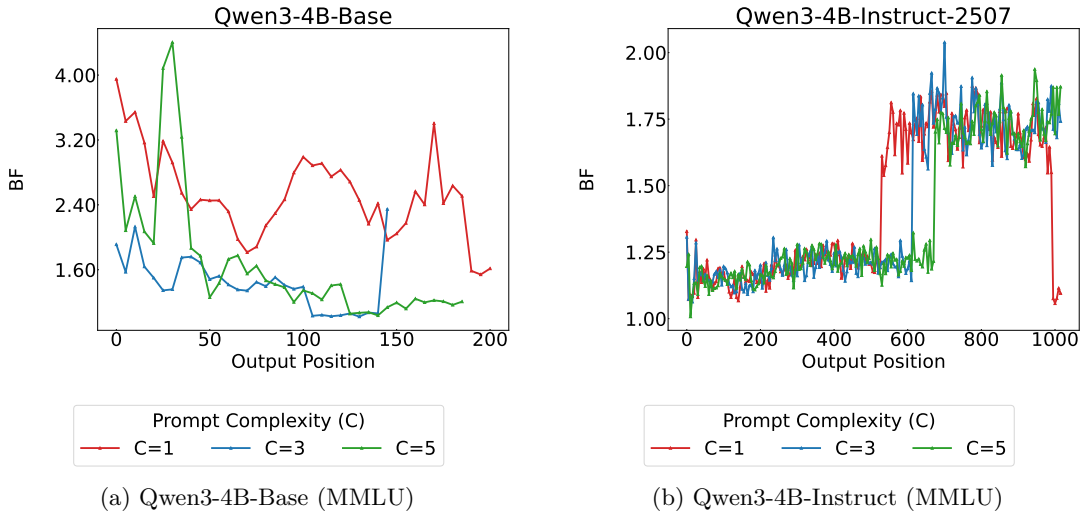


Figure 15: **BF Output Dynamic for Qwen3-4B on MMLU.**